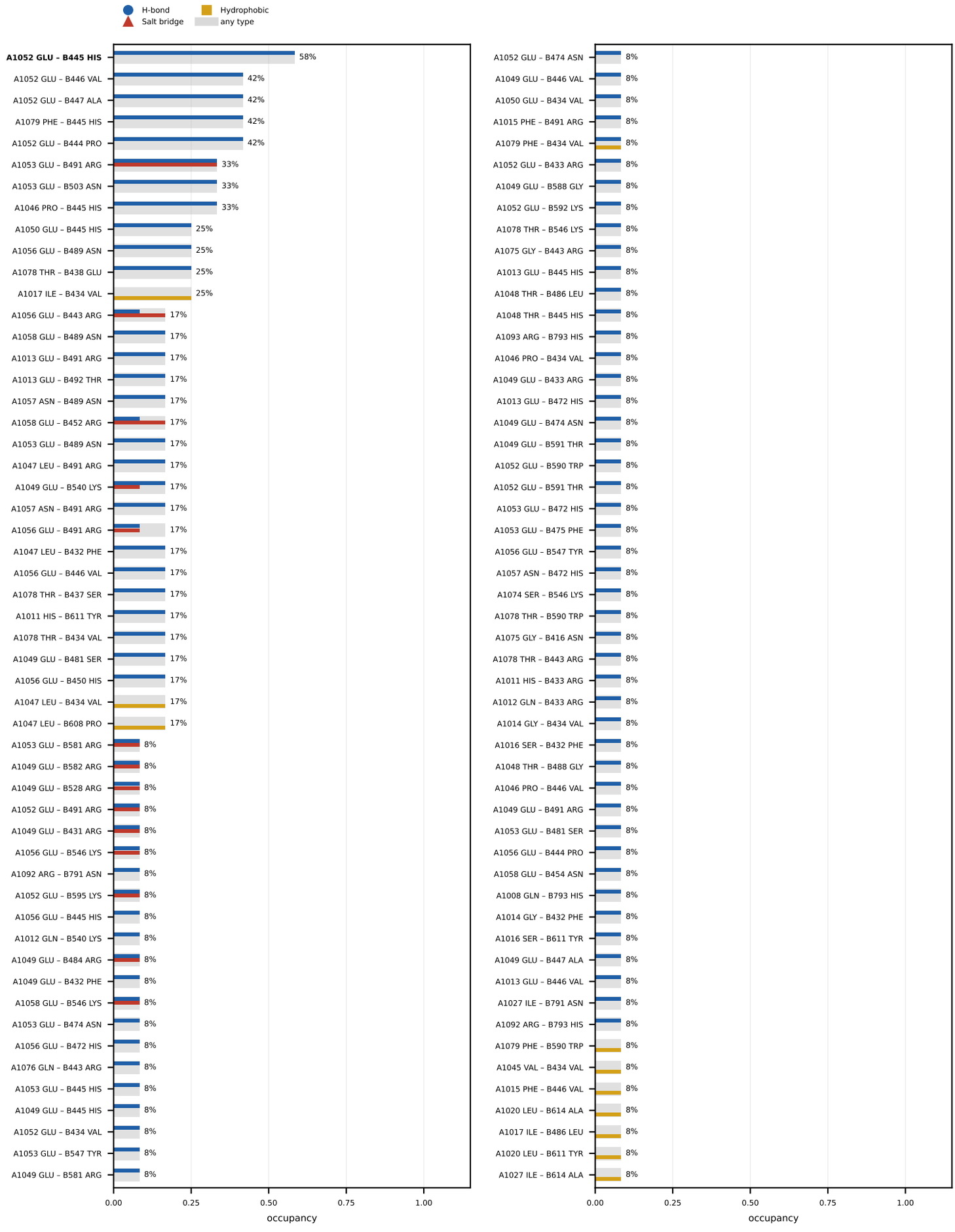E. All residue pairs (12 models, n=106)